

복사전달 특강

Special Topics in Radiative Transfer

Lecture 15
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선광일 (Kwangil Seon)
KASI / UST

Outline – MCRT and Dynamics

Why Couple Radiation and Hydrodynamics?

The Radiation-Hydrodynamics Equations

Reconstructing the Transfer Terms from MC Packets

Coupling to Fluid Dynamics: Operator Splitting

A Standard Test: Radiative Shocks

Stiff Source Terms and Limitations

Summary

Radiation carries energy *and* momentum

Everything so far treated the medium as *fixed*: given a density, temperature, and velocity field, we solved for the radiation field. But a photon of energy $h\nu$ also carries momentum $h\nu/c$. Each absorption or scattering event therefore deposits both:

- ▶ **Energy** $\rightarrow$ heats (or cools) the gas — “radiative heating.”
- ▶ **Momentum** $\rightarrow$ pushes the gas — the “radiation force.”

When these are strong enough to *change the gas motion*, the density and velocity field that shaped the radiation is itself reshaped by the radiation. The two can no longer be solved separately:

$$\boxed{\text{radiation field} \rightleftharpoons \text{gas dynamics}}$$

This coupled problem is **radiation hydrodynamics** (RHD, or “RH”). This lecture asks: *how do we feed a Monte Carlo radiative-transfer (MCRT) calculation into a hydrodynamics code?*

Where radiation dynamics matter

A short tour of regimes where the radiation force is not a small correction but the *driver* of the dynamics:

- ▶ **Line-driven winds from hot, massive stars.** UV photons resonantly scatter off thousands of metal lines; the cumulative momentum exceeds gravity and launches a supersonic wind (CAK theory — Castor, Abbott & Klein 1975; MC treatment — Abbott & Lucy 1985).
- ▶ **Accretion-disc winds** in AGN, CVs, and X-ray binaries, launched by line and continuum radiation pressure (Proga et al. 2000).
- ▶ **Massive star formation.** Dust reprocesses the stellar luminosity; the dust radiation force can halt infall and help set the final stellar mass (Harries 2015; Nayakshin et al. 2009).
- ▶ **Supernovae:** shock breakout, and radiation-dominated ejecta where $E_{\text{rad}} \gg e_{\text{gas}}$ and radiation pressure governs the dynamics (overview in Mihalas & Mihalas 1984).

Why Monte Carlo for these problems?

Historically RHD was solved with *deterministic* (grid-based moment) methods. Monte Carlo became attractive for two reasons:

- **Sobolev line opacities are trivial to sample.** In a fast-moving medium a photon resonates with a given line only in a thin *Sobolev layer* where its comoving frequency crosses line center (Sobolev 1960). The interaction probability is the local algebraic factor $1 - e^{-\tau_s}$ with

$$\tau_s = \frac{\pi e^2}{m_e c} f_{lu} n_\ell \lambda_0 \left(\frac{d}{ds} \right)^{-1}.$$

A packet simply hops from one resonance point to the next; multi-line and line-overlap effects come for free.

- **Indivisible energy packets** (Lucy 1999) conserve energy *by construction*: absorption is followed by immediate re-emission of an equal-energy packet, so a single packet supplies both the absorption *and* the re-emission parts of the force estimator.

Production codes: MCRH (Noebauer et al. 2012), Roth & Kasen (2015), TARDIS (SN Ia spectra), TORUS (star formation).

The Sobolev picture: one resonance at a time

Why is line transport in a *moving* medium so cheap? A large velocity gradient makes each line resonate only in a *thin layer*. A packet of lab-frame frequency ν travelling along $\mathbf{n}$ has, to $O(1/c)$, the comoving frequency

$$\nu_{\text{cmf}}(s) = \nu \left[1 - \frac{\mathbf{n} \cdot \mathbf{u}(s)}{c} \right],$$

which *drifts* as the projected velocity $\mathbf{n} \cdot \mathbf{u}$ changes along the ray — redshifting in an expanding flow. The packet meets a line of rest frequency ν_0 only at the **Sobolev point** s_S where $\nu_{\text{cmf}}(s_S) = \nu_0$. Across that thin layer n_ℓ and $\mathbf{u}$ are constant and only ν_{cmf} varies, shifting through the profile at $d\nu_{\text{cmf}}/ds = -(\nu_0/c) d/ds$, so the opacity integral collapses to a purely *local* algebraic factor (Sobolev 1960):

$$\tau_S = \int \chi_\nu ds = \frac{\pi e^2}{m_e c} f_{\ell u} n_\ell \lambda_0 \left(\frac{d}{ds} \right)^{-1}, \quad p_{\text{int}} = 1 - e^{-\tau_S}.$$

- ▶ $\frac{\pi e^2}{m_e c}$ — classical line-absorption constant; $f_{\ell u}$ — oscillator strength; n_ℓ — lower-level number density; $\lambda_0 = c/\nu_0$.
- ▶ $(d/ds)^{-1}$ — the velocity gradient $d/ds = \mathbf{n} \cdot (\mathbf{n} \cdot \nabla) \mathbf{u}$. A *steeper* gradient sweeps the photon through the line faster $\Rightarrow$ thinner layer $\Rightarrow$ smaller τ_S .

Multi-line transfer and line overlap

Multi-line — for free. Because v_{cmf} drifts *monotonically*, a single packet meets a whole *sequence* of lines, encountered in order of rest frequency. The transport is a simple loop:

1. find the next line the drifting v_{cmf} will reach;
2. propagate the packet to that Sobolev point s_S ;
3. interact with probability $1 - e^{-\tau_S}$ (on interaction, re-emit at line center in the CMF, in a new random direction); otherwise pass through untouched;
4. repeat for the next line.

A list of *thousands* of metal lines (CAK line-driven winds; SN Ia spectra) costs just *one stop per line* — no overlapping-profile integral, no global frequency grid.

Line overlap. Two lines of rest frequencies v_1, v_2 resonate at points separated by

$$\Delta s \simeq \frac{c |v_1 - v_2|}{v_0} \left(\frac{d}{ds} \right)^{-1}.$$

As long as Δs exceeds the (tiny) intrinsic profile width, the packet handles the lines *sequentially and independently*. Crucially, *non-local* line coupling emerges automatically: a photon scattered by line 1 simply keeps drifting and may be re-absorbed by line 2 downstream. When the resonance regions truly coincide, the joint escape uses the summed depth $\tau_{\text{tot}} = \sum_k \tau_{S,k}$ (multiline transfer — Abbott & Lucy 1985).

Euler equations + a radiation source term

The fluid still obeys conservation of mass, momentum, and energy — but the momentum and energy equations gain a *radiation four-force* on the right-hand side (Mihalas & Auer 2001; chapter Eqs. 143–145):

$$\frac{\partial \rho}{\partial t} + \sum_i \frac{\partial (\rho^i)}{\partial x^i} = 0, \quad (\text{mass — unchanged})$$

$$\frac{\partial (\rho^i)}{\partial t} + \sum_j \frac{\partial}{\partial x^j} (\rho^{ij} + p \delta^{ij}) = G^i - \frac{i}{c} G^0, \quad (\text{momentum})$$

$$\frac{\partial}{\partial t} \left[\rho \left(\frac{1}{2}^2 + e \right) \right] + \sum_i \frac{\partial}{\partial x^i} \left[\rho^i \left(\frac{1}{2}^2 + e \right) + p^i \right] = c G^0. \quad (\text{energy})$$

Here ρ is density, p pressure, e specific internal energy.

- ▶ G^0 — **energy** exchange (heating/cooling).
- ▶ G^i — **momentum** exchange (the radiation force).

Set $G^0 = G^i = 0$ and we recover the ordinary Euler equations.

What is the radiation four-force?

The four-force is just “net emitted minus net absorbed,” integrated over all frequencies and all directions (chapter Eqs. 146–147):

$$\begin{aligned} -c G^0 &= \int_0^\infty d\nu \int d\Omega (\eta_\nu - \chi_\nu I_\nu), \\ -c G^i &= \int_0^\infty d\nu \int d\Omega (\eta_\nu - \chi_\nu I_\nu) n^i. \end{aligned}$$

- ▶ χ_ν — opacity (extinction per unit length); η_ν — emissivity; I_ν — specific intensity.
- ▶ $\chi_\nu I_\nu$ is energy *removed* from the beam; η_ν is energy *added*. The direction cosine n^i in G^i projects the photon momentum onto axis i .
- ▶ **Action–reaction:** whatever the radiation loses, the gas gains. That is why the same G^0, G^i appear (with opposite sign) in both the fluid and radiation energy budgets.

Lab frame vs. comoving frame

Two frames, each natural for one half of the problem:

- ▶ **Comoving frame (CMF, subscript 0)** — the local rest frame of the gas. Atomic cross sections, the Planck function $B_{\nu}(T)$, and LTE (Kirchhoff: $\eta_{0\nu} = \chi_{0\nu} B_{\nu}$) are *isotropic and simple* here. Opacities are defined in the CMF.
- ▶ **Lab frame (LF)** — the fixed grid. Photons *propagate* here.

We transform between them to $O(v/c)$ (or to all orders). A key consequence: an *isotropic* comoving opacity becomes *direction-dependent* in the lab frame — photons moving *with* the flow have longer mean free paths than those moving *against* it. This is exactly what produces the correct **advection of radiation** by the moving gas.

In LTE the comoving radiation energy density obeys Stefan–Boltzmann, $E_0 = a_R T^4$, with the radiation constant $a_R = 4\sigma_{\text{SB}}/c \approx 7.566 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$. This sets the thermal-emission term.

Direction-dependent opacity $\Rightarrow$ radiation advection

Unpack that last claim: a lab-frame photon ($\nu, \mathbf{n}$) has, in the gas rest frame, the Doppler-shifted frequency

$$\nu_0 = \nu \left(1 - \frac{\mathbf{n} \cdot \mathbf{u}}{c} \right) \quad [O(/c)].$$

The opacity is isotropic and simple only *in the CMF*, $\chi_0(\nu_0)$. Transforming the extinction coefficient to the lab frame multiplies it by the Doppler factor ν_0/ν ; for a grey opacity this leaves an explicit *direction* dependence (Mihalas & Mihalas 1984):

$$\boxed{\chi(\mathbf{n}) = \chi_0 \left(1 - \frac{\mathbf{n} \cdot \mathbf{u}}{c} \right)} \quad \Rightarrow \quad \ell(\mathbf{n}) = \frac{1}{\chi(\mathbf{n})} = \ell_0 \left(1 + \frac{\mathbf{n} \cdot \mathbf{u}}{c} \right).$$

- **With the flow** ($\mathbf{n} \cdot \mathbf{u} > 0$): the photon sees gas *receding*, opacity drops, mean free path ℓ is *longer*.
- **Against the flow** ($\mathbf{n} \cdot \mathbf{u} < 0$): gas *approaching*, opacity rises, ℓ is *shorter*.

So a source that is *isotropic* in the comoving frame still leaks preferentially *downstream* — a net lab-frame flux builds up along $\mathbf{u}$. Summing the first-order anisotropy over all directions, the moment transformation gives

$$\mathbf{F} = \mathbf{F}_0 + (E_0 + P_0) \mathbf{u} \xrightarrow{P_0=E_0/3} \mathbf{F}_0 + \frac{4}{3} E_0 \mathbf{u},$$

i.e. the comoving flux *plus* the radiation enthalpy $\frac{4}{3} E_0 \mathbf{u}$ dragged along by the gas. That extra term *is* the **advection of radiation** by the moving fluid.

The grey limit: clean, intuitive source terms

Define the angular moments of the radiation field

$$E = \frac{1}{c} \oint I d\Omega, \quad F^i = \oint I n^i d\Omega, \quad P^{ij} = \frac{1}{c} \oint I n^i n^j d\Omega$$

(energy density, flux, pressure tensor). For **grey** opacity (frequency-independent), the comoving source terms collapse to the transparent forms (chapter Eqs. 150–151):

$$\boxed{c G_0^0 = c \chi_{a,0} (E_0 - a_R T_0^4)} \quad \boxed{c G_0^i = \chi_{\text{tot},0} F_0^i}$$

- **Energy:** the gas is heated if the local radiation energy density E_0 exceeds the equilibrium value $a_R T_0^4$, and cools otherwise — it relaxes toward radiative equilibrium. $\chi_{a,0}$ is the *absorption* opacity.
- **Momentum:** the force is the total opacity times the radiative flux. A net flux is needed to push the gas.

For non-grey opacities one introduces frequency-averaged opacities (Planck mean for emission, flux mean for the force; Roth & Kasen 2015).

Volume-based (path-length) estimators

How do we measure E_0, F_0^i, G^0, G^i from a finite set of noisy packet histories? The naive way — tally only packets that actually *interact* in a cell — wastes almost all the information and is very noisy.

Lucy's (1999) trick. Tally *every segment* of *every* trajectory that crosses the cell, weighted by the path length l (i.e. by the time the packet spent there):

$$E = \frac{1}{\Delta V c \Delta t} \sum_i \varepsilon l_i, \quad J = \frac{c}{4\pi} E,$$

where ε is the packet energy and the sum runs over all segments l_i in cell volume ΔV during time step Δt .

Every packet contributes to every cell it merely *passes through* — not just where it interacts. The estimator uses the *probability* that an interaction occurs, so the variance is dramatically lower. This is the same idea used for heating rates and mean intensities throughout the review.

Force estimators

Weighting each segment by the opacity and the appropriate direction cosine μ_0 gives the comoving energy- and momentum-transfer terms directly (chapter Eqs. 152–153):

$$G_0^0 = \frac{1}{\Delta V c \Delta t} \sum \chi_{a,0} l_0 \epsilon_0 - 4\pi \int_0^\infty dv_0 \chi_{a,0}(v_0) B_{v_0},$$

$$G_0^i = \frac{1}{\Delta V c \Delta t} \sum \chi_{\text{tot},0} l_0 \mu_0 \epsilon_0.$$

- ▶ The sum is “packet energy $\times$ probability of interacting along the segment l_0 ,” accumulated over the whole packet ensemble.
- ▶ The subtracted integral in G_0^0 is the gas’s own *thermal emission* ($\propto \int \chi_{a,0} B_{v_0}$): net heating = absorbed – emitted.
- ▶ The opacity is evaluated at the packet’s instantaneous frequency at the *start* of each segment — valid when the opacity varies only mildly along l_0 . Strong line resonances (SN Ia, hot-star winds) need the special treatments of Lucy (1999) / Noebauer & Sim (2015).

The diffusion-regime trap — and a clever fix

The problem. The momentum estimator G_0^i depends on the *net* flux F_0^i . But the flux is the small difference of large, opposing contributions from packets going one way and the other:

$$F_0^i = \underbrace{(\text{forward})}_{\text{large}} - \underbrace{(\text{backward})}_{\text{large}}.$$

In the **diffusion regime** (optically thick) the field is nearly isotropic, so F_0^i is a tiny residual — and the Monte Carlo noise on the two large sums does *not* cancel. The signal-to-noise of the force collapses (Roth & Kasen 2015).

The fix (Roth & Kasen 2015). Under diffusive conditions the radiation momentum equation reduces (Mihalas & Auer 2001) to

$$G^i = - \sum_j \frac{\partial P^{ij}}{\partial x^j}$$

Now the force is the *divergence of the radiation pressure tensor*. P^{ij} is tallied by a *quadratic* path-length sum $\propto \sum \epsilon \ln^i n^j$ that involves *no cancellation* (every term is positive-definite in the diagonal), so it converges cleanly even deep in the diffusion regime.

Operator splitting in one slide

We now have G^0, G^i . How do we advance the *coupled* system in time? Almost universally by **operator splitting** (LeVeque 2002): solve each physical process *separately* over a time step rather than the full coupled system at once.

For $\partial_t U = \mathcal{A}(U) + \mathcal{B}(U)$:

- ▶ **Godunov (Lie) splitting:** apply $\mathcal{A}$ for a full step, then $\mathcal{B}$ for a full step. First-order accurate in Δt . Simplest; used by MCRH.
- ▶ **Strang splitting:** half-step $\mathcal{A}$, full step $\mathcal{B}$, half-step $\mathcal{A}$ ($\mathcal{A}^{\Delta t/2} \mathcal{B}^{\Delta t} \mathcal{A}^{\Delta t/2}$). Second-order accurate, one extra sub-step.

This is the standard way to treat *source terms* (gravity, nuclear burning, radiation) in hydrodynamics codes. The radiation step is treated as a source term that updates the fluid momentum and energy.

The radiation-hydrodynamics time-step loop

Each step: hydro alone $\rightarrow$ MC transfer $\rightarrow$ apply the radiation source term $\rightarrow$ repeat.

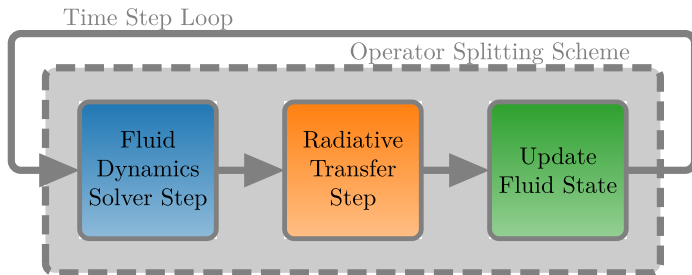


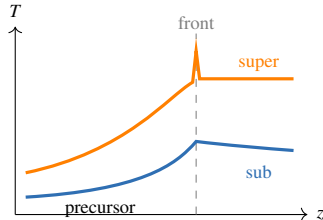
Figure 1: Godunov operator-splitting scheme for MC-based radiation hydrodynamics. Reproduced from Noebauer & Sim (2019), Fig. 17.

1. **Fluid step:** advance the Euler equations with a Godunov/Riemann solver, ignoring radiation ($G^0=G^i=0$) $\rightarrow$ new $\rho, \mathbf{u}, e$.
2. **RT step:** propagate MC packets through the *frozen* new fluid; accumulate the path-length estimators for G^0, G^i .
3. **Update fluid state:** apply the source terms $\rho \partial_t \mathbf{u} = G^i$ and $\rho \partial_t e = cG^0$. The step is complete.

Radiative shocks and the radiative precursor

Drive gas into a wall: a shock forms. If the post-shock gas is hot enough to radiate strongly, photons stream *upstream* and pre-heat the gas *ahead* of the front — a **radiative precursor**. Two regimes:

- ▶ **Sub-critical:** weak precursor; pre-shock temperature T_- stays *below* the post-shock T_+ .
- ▶ **Super-critical:** strong precursor; gas just ahead of the front reaches T_+ . The jump is washed out and a narrow *Zel'dovich spike* sits at the front. (T_- can never exceed T_+ .)



Solid curves: gas temperature. Upstream is to the left of the front, shocked gas to the right. Schematic after Ensman (1994) and Noebauer et al. (2012).

Benchmark and results

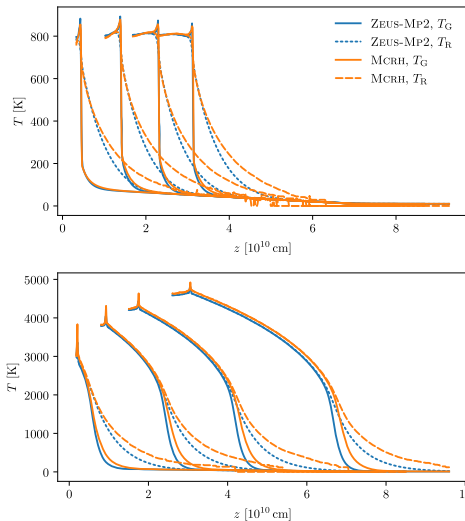
The test problem. Proposed by Ensman (1994); now the standard RHD benchmark. Gas is driven into a reflecting boundary (e.g. -6 km s^{-1} sub-critical, -20 km s^{-1} super-critical). Control parameters: the piston Mach number (shock strength) and the resulting gas temperature T_g vs. radiation temperature T_R profiles.

Reference solutions.

- ▶ Lowrie & Rauenzahn (2007): steady solutions in the *equilibrium*-diffusion limit ($T_g = T_R$).
- ▶ Lowrie & Edwards (2008): grey *non*-equilibrium diffusion (separate T_g, T_R) — the more complete benchmark, e.g. a steady Mach $M = 5$ shock.

Outcome. Both MCRH (Noebauer et al. 2012) and Roth & Kasen (2015) reproduce sub- and super-critical shocks. The Monte Carlo runs agree very well with finite-difference (ZEUS-MP2) and with the semi-analytic Mach-5 solution — validating the whole estimator + operator-splitting machinery.

Non-steady shocks: Monte Carlo vs. finite difference



Temperature structure at four times.

Top: sub-critical shock.

Bottom: super-critical shock.

Orange: MCRH (Monte Carlo).

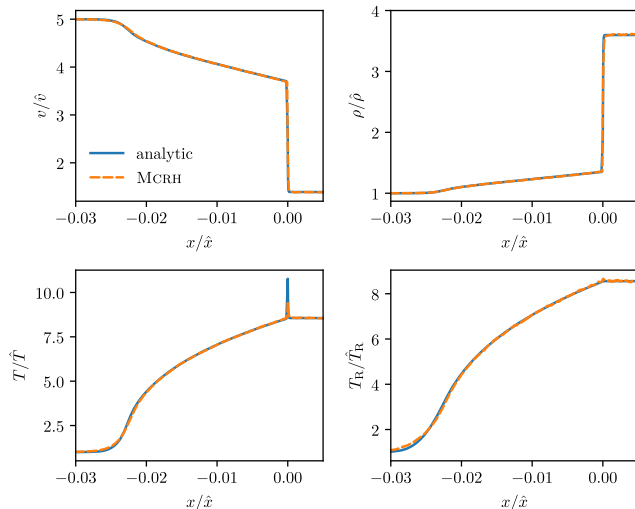
Blue: ZEUS-MP2 (finite difference).

Solid = gas T_G , dashed/dotted = radiation T_R .

The narrow *Zel'dovich spike* sits right at the front in the super-critical case (bottom). The two independent codes track each other closely.

Adapted from Noebauer & Sim (2019), Fig. 18 (after Noebauer et al. 2012).

Steady Mach-5 shock: Monte Carlo vs. analytic



A steady radiative shock with Mach number $M = 5$, all quantities in non-dimensional form vs. position $x/\hat{x}$.

Blue solid: semi-analytic solution (Lowrie & Edwards 2008).

Orange dashed: MCRH.

Clockwise from top left: velocity v , density ρ , radiation temperature T_R , gas temperature T .

Note the smooth *radiative precursor* upstream ($x < 0$) and the sharp *Zel'dovich spike* in the gas temperature at the front ($x = 0$). The Monte Carlo result reproduces the analytic profile essentially exactly.

Adapted from Noebauer & Sim (2019), Fig. 19.

The stiffness problem and the cooling time

Operator splitting is conceptually simple but has a sharp limitation: the radiation source terms carry their *own* timescale — the radiative cooling time (chapter Eq. 155; to $O(/c)$):

$$\boxed{t_{\text{cool}} = \frac{\rho e}{c G^0}} \quad (\text{gas energy content / radiative loss rate}).$$

The explicit hydro step, meanwhile, is limited by the Courant–Friedrichs–Lewy (CFL) condition,

$$\Delta t_{\text{CFL}} \lesssim \frac{\Delta x}{|| + c_s}.$$

The trap. If $t_{\text{cool}} \ll \Delta t_{\text{CFL}}$, applying the source term explicitly over a full step *over-cools* the cell: thermal emission drains the entire internal-energy budget and drives $e < 0$ — an unphysical state, and the code crashes. The source terms are then said to be **stiff**. This happens whenever radiative timescales are much shorter than fluid-flow timescales (radiation-dominated regimes).

The cooling time, unpacked

Where does t_{cool} come from? Strip the hydro terms from the gas energy equation and keep only the radiation source acting on the *internal* energy — exactly the update applied each step ($\rho \partial_t e = c G^0$):

$$\frac{d(\rho e)}{dt} = c G^0.$$

Here ρe is the thermal energy stored *per unit volume* and $c G^0$ is the net radiative exchange rate per unit volume (absorbed – emitted). When the cell is a net *emitter*, $G^0 < 0$, so ρe decays; dividing the content by the rate gives the characteristic e-folding time of that decay,

$$\boxed{t_{\text{cool}} = \frac{\rho e}{|c G^0|}} = \frac{\text{thermal energy stored}}{\text{rate it is radiated away}}.$$

Make it concrete with the grey source term $c G_0^0 = c \chi_{a,0} (E_0 - a_R T^4)$ (this lecture, grey limit). In a transparent, strongly emitting cell ($E_0 \ll a_R T^4$) the ambient term drops and, using $\rho e = \frac{3}{2} n k_B T$ for an ideal gas,

$$|c G^0| \simeq c \chi_{a,0} a_R T^4 \quad \Longrightarrow \quad t_{\text{cool}} \simeq \frac{\rho e}{c \chi_{a,0} a_R T^4} = \frac{3 n k_B}{2 c \chi_{a,0} a_R} T^{-3}.$$

The steep $t_{\text{cool}} \propto T^{-3}$ is the heart of the stiffness: a hot cell radiates its thermal energy almost instantaneously, so t_{cool} easily drops far below Δt_{CFL} .

Remedies for stiff source terms

1. Implicit Monte Carlo (IMC; Fleck & Cummings 1971). Linearize the matter–radiation coupling over the step and recast a fraction of true absorption as instantaneous, isotropic re-emission (“*effective scattering*”). The *Fleck factor*

$$f = \frac{1}{1 + \alpha (c \Delta t \chi_P) a_R T^4 / (\rho e)}, \quad \alpha \in [\tfrac{1}{2}, 1],$$

interpolates: $f \rightarrow 1$ (explicit) when $\Delta t \ll t_{\text{cool}}$, and $f \ll 1$ (mostly effective scattering, bounded energy update) when $\Delta t \gg t_{\text{cool}}$. Roth & Kasen (2015) coupled IMC with hydrodynamics, allowing $\Delta t \gg t_{\text{cool}}$.

Caveat. IMC is only *semi*-implicit (opacities are lagged): for very large Δt it can violate the maximum principle (temperature overshoots) and show teleportation errors (Wollaber 2016).

2. Unsplit, semi-implicit Godunov schemes. Treat the stiff source term *inside* the Godunov update implicitly (Miniati & Colella 2007; Sekora & Stone 2010; Jiang et al. 2012). Robust across the stiff regime; not yet demonstrated with full MCRT.

Take-home messages

The coupling. Radiation deposits energy (G^0) and momentum (G^i) into the gas. When these reshape the flow, radiation and hydrodynamics must be solved together — radiation hydrodynamics.

The MC recipe. Reconstruct G^0, G^i from path-length (volume-based) estimators summed over *all* packet trajectory segments — low noise, every passing packet counts. In the diffusion regime, reconstruct the force from the divergence of the pressure tensor P^{ij} to dodge the flux-cancellation noise (Roth & Kasen 2015).

The coupling loop. Operator (Godunov) splitting: hydro step $\rightarrow$ MC transfer step $\rightarrow$ apply the radiation source term $\rightarrow$ repeat. Validated on sub- and super-critical radiative shocks.

The catch. Explicit splitting fails when $t_{\text{cool}} \ll \Delta t_{\text{CFL}}$ (stiff source terms, negative energies). Implicit Monte Carlo and unsplit semi-implicit Godunov schemes are the cures — at the cost of new accuracy caveats (maximum-principle violations).

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